Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1

Answer QUESTION 1 PARTS A and B on this page.

PART A

$$\frac{1}{4-0}\int_0^4 C(t)\,dt \approx 2.778 \text{ acres}$$

PART B

$$\frac{C(4)-C(0)}{4-0} \approx 1.282 \text{ acres/week}$$

$$C'(t) = \frac{38}{25+t^2} = 1.282 \text{ at } t = 2.154$$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1 Q1

Answer QUESTION 1 PARTS C and D on this page.

PART C

$$\lim_{t\to\infty} C'(t) = \lim_{t\to\infty} \frac{38}{25+t^2} = 0$$

PART D

$A'(t) = 0$ at $t = 11.442$

t	$A(t)$
4	5.128
11.442	7.317 ← largest
36	1.743

A attains its maximum value at $t = 11.442$ for $4 \le t \le 36$.

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